Assignment 3 Report
Financial Engineering

Group 5

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Abstract

This report presents the results for Assignment 3 of the Financial Engineering course. We cover Gaussian parametric VaR/ES (Exercise 0), Historical Simulation, Bootstrap, WHS and PCA-based risk measures (Exercise 1), Full Monte-Carlo and Delta-Normal VaR (Exercise 2), Cliquet option pricing under counterparty risk (Exercise 3), and MBS tranche pricing under the Vasicek model (Exercise 4).

Exercise 0: Variance-Covariance VaR & ES

We compute the 1-day VaR and ES at a 95% confidence level. The portfolio is equally weighted with Inditex, BASF, and LVMH. The notional is €10 Mln, the reference date is July 24, 2012, and we use a 2-year historical data window.

We apply a Gaussian parametric approach. Portfolio losses are modeled as $L_t = -\mathbf{w}^\top \mathbf{r}_t \sim \mathcal{N}(\mu_L, \sigma_L^2)$. Therefore, we derive VaR and ES directly using the standard normal quantile and probability density functions.

To manage missing trading days across different exchanges, we forward-fill the data using the previous day's price.

Measure	**Value (€)**
VaR (95%)	244,867
ES (95%)	308,388

Table 1: Exercise 0: Gaussian VaR and ES.

The Expected Shortfall is about 26% higher than the VaR. This relationship is exactly what we expect from a Normal distribution, where the theoretical ES/VaR ratio at a 95% confidence level is roughly 1.26.

Overall, these values are highly realistic for a €10 Mln equity portfolio. The daily VaR represents approximately 2.45% of the total notional, which is a standard and plausible risk level for a moderately diversified portfolio of three large-cap stocks.

Exercise 1: HS, Bootstrap, WHS and PCA Risk Measures

All portfolios are evaluated as of 24/07/2012 with $\alpha = 99\%$ and a 2-year estimation window.

Exercise 1a: Historical Simulation & Bootstrap

Portfolio 1 consists of Eni (18K shares), Telefonica (25K), Eon (15K), and Daimler (9K). Portfolio weights are computed from current market prices.

Method	**VaR (€)**	**ES (€)**
Historical Simulation	56,147	63,031
Bootstrap ($M = 200$)	51,406	60,148
Plausibility Check	46,721	–

Table 2: Exercise 1a: VaR and ES for Portfolio 1.

The Historical Simulation and Bootstrap VaR are very close. The Bootstrap estimate is slightly lower because averaging 200 resampled scenarios reduces the impact of the extreme tail losses. Finally, both values align well with the plausibility check, confirming that the results are realistic.

Exercise 1b: Weighted Historical Simulation

Portfolio 2 is equally weighted across Vivendi, AXA, Enel, Volkswagen, and Schneider. We use $\lambda = 0.98$ and normalize portfolio value to 1 for percentage-based reporting.

Method	**VaR**	**ES**
WHS ($\lambda = 0.98$)	4.14%	4.96%
Plausibility Check	5.06%	–

Table 3: Exercise 1b: VaR and ES for Portfolio 2 (as % of notional).

The WHS VaR is lower than the Plausibility Check. The Plausibility Check provides a baseline Variance-Covariance estimate by weighting the entire 2-year historical window equally. In contrast, the WHS method applies an exponential decay ($\lambda = 0.98$), assigning much higher weights to the most recent observations. Since the market in mid-2012 was less volatile than the peak of the 2010-2011 European debt crisis, focusing heavily on recent data naturally leads to a lower risk estimate.

Exercise 1c: PCA-Based Gaussian VaR & ES

Portfolio 3 is equally weighted across 25 EUROSTOXX 50 constituents, with €15 Mln notional and a 10-day horizon.

Measure	**Value (€)**
VaR (PCA, full $k = 25$)	1,849,692
Plausibility Check	2,039,474
Min. k for $< 5\%$ error	1
Min. k for $< 1\%$ error	1

Table 4: Exercise 1c: PCA-based VaR for Portfolio 3.

A single principal component ($k = 1$) keeps the error below 1%. This means the first component drives almost all the portfolio's variance, reflecting the dominant "market factor" typical of well-diversified portfolios.

The full PCA VaR is slightly lower than the Plausibility Check. This makes perfect theoretical sense. The Plausibility Check is an elliptical approximation: it extracts the empirical quantiles of individual stocks (capturing actual fat tails) and aggregates them via the correlation matrix. The PCA, instead, relies on a strict Gaussian assumption, which slightly underestimates extreme events.

Despite this technical difference, the two values are close. This confirms that our VaR estimates are robust and the elliptical approximation is reliable.

Exercise 2: Full Monte-Carlo and Delta-Normal VaR

As of 15/02/2010, the portfolio consists of €1,164,000 invested in Generali shares and the same number of European put options (strike 28.5 €, expiry 18/04/2010, $\sigma = 22.3\%$, $q = 5.1\%$). We compute 1-day VaR at $\alpha = 99\%$.

We evaluate the portfolio loss using two different approaches:

- **Full Monte-Carlo:** This method computes the exact loss. It fully reprices the derivatives for each simulated scenario:

$$L = -\sum [\mathcal{C}_i(t + \Delta) - \mathcal{C}_i(t)]$$

- **Delta-Normal:** This method uses a first-order linear approximation. It computes the loss using the portfolio sensitivity (Delta) and the risk-factor variations (X_t):

$$L(X_t) = -\sum_{i=1}^{d} \text{sens}_i(t) X_{t;i}$$

For both methods, the risk-factor variations are generated using a 2-year Historical Simulation.

Method	**VaR (€)**
Full Monte-Carlo	506
Delta-Normal	565

Table 5: Exercise 2: VaR comparison.

A VaR of approximately 500 € on a 1.16 million € portfolio is exceptionally low, primarily because the portfolio is almost perfectly hedged. The current stock price (approx. 16.5 €) is far below the put option strike (28.5 €). This means the put option is deep in-the-money. The stock has a Delta of +1, while the deep in-the-money put has a Delta of almost -1. The net Delta of the entire portfolio is close to zero (Delta-neutral). Any loss from a drop in the stock price is almost entirely offset by an equal gain in the put option.

Furthermore, the discrepancy between the two methods is very small, with the Delta-Normal VaR slightly overestimating the risk by a minimal amount (around 59 €). This happens because a deep in-the-money option has a Gamma very close to zero. Gamma measures the non-linear curvature of the option's price. Since the curvature is negligible, the linear approximation (Delta-Normal) performs remarkably well and behaves almost exactly like the exact repricing model (Full Monte-Carlo). The small difference exists because the tiny residual positive Gamma still provides a slight non-linear benefit during market drops, which the purely linear model ignores.

Can you improve the Delta-Normal VaR? How? Yes, it can be improved by using the Delta-Gamma approach. This method includes the second-order Taylor expansion term (the

Gamma). By incorporating Gamma, the analytical model captures the exact curvature of the portfolio. This correction reduces the estimated VaR and aligns it with the exact Full Monte-Carlo result.

Why can Full Monte-Carlo be numerically intensive for an exotic derivative? Full Monte-Carlo requires a complete repricing of the derivative for every single simulated scenario. Exotic derivatives usually lack closed-form pricing formulas (like the standard Black-Scholes formula). To find the price in each scenario, you must run an additional numerical simulation. This creates a nested simulation structure (a "Monte-Carlo within a Monte-Carlo"), making the computational time extremely high.

Exercise 3: Pricing in Presence of Counterparty Risk

On 15/02/2008, bank XX buys from ISP a 5-year Cliquet option (€45 Mln notional) with yearly payoff $[S(t_i) - S(t_{i-1})]^+$. The underlying has no dividends and $\sigma = 19\%$.

Each annual payoff is an ATM call on the underlying's return. To derive the total price, we evaluate the net present value (NPV) of the 5 single yearly payoff under the risk-neutral measure $\mathbb{Q}$ using the following steps:

1. **Present Value of a single annual payoff:** For any year $i \in \{1, \dots, 5\}$, the value at t_0 is defined as:
$$NPV_i = e^{-rt_i}\mathbb{E}^{\mathbb{Q}}\left[(S(t_i) - S(t_{i-1}))^+\right]$$

2. **Conditioning on $S(t_{i-1})$:** Using the Law of Iterated Expectations, we condition the payoff on the spot price at the start of the period, $S(t_{i-1})$:
$$NPV_i = e^{-rt_{i-1}}\mathbb{E}^{\mathbb{Q}}\left[e^{-r(t_i - t_{i-1})}\mathbb{E}^{\mathbb{Q}}[(S(t_i) - S(t_{i-1}))^+ \mid S(t_{i-1})]\right]$$
The inner expectation $\mathbb{E}^{\mathbb{Q}}[\cdots \mid S(t_{i-1})]$ represents the Black–Scholes price of an ATM call option starting at t_{i-1}.

3. **Scaling Property of Black–Scholes:** By the linear homogeneity of the Black–Scholes formula $C_{\text{BS}}(S, K, r, \sigma, T)$, where $K = S(t_{i-1})$ and $T = t_i - t_{i-1} = 1$, we can scale the spot price out:
$$C_{\text{BS}}(S(t_{i-1}), S(t_{i-1}), r, \sigma, 1) = S(t_{i-1}) \cdot C_{\text{BS}}(1, 1, r, \sigma, 1)$$

4. **Martingale Property and Simplification:** Substituting this into the NPV_i formula and moving the constant term $C_{\text{BS}}(1, 1, r, \sigma, 1)$ outside the expectation:
$$NPV_i = e^{-rt_{i-1}} \cdot C_{\text{BS}}(1, 1, r, \sigma, 1) \cdot \mathbb{E}^{\mathbb{Q}}[S(t_{i-1})]$$
Given the absence of dividends, $\mathbb{E}^{\mathbb{Q}}[S(t_{i-1})] = S_0 e^{rt_{i-1}}$. Therefore:
$$NPV_i = e^{-rt_{i-1}} \cdot C_{\text{BS}}(1, 1, r, \sigma, 1) \cdot S_0 e^{rt_{i-1}} = S_0 \cdot C_{\text{BS}}(1, 1, r, \sigma, 1)$$

Since each coupon has the same net present value at t_0, the total price for a 5-year Cliquet option with Notional N is:

$$\text{Price} = N \cdot 5 \cdot S_0 \cdot C_{\text{BS}}(1, 1, r, \sigma, 1)$$

Quantity	**Value**
Risk-free rate (5y)	3.97%
Cliquet price (risk-free)	€21,434,141

Table 6: Exercise 3: Cliquet option price.

The risk-free price of the Cliquet is approximately €21.4 Mln, which corresponds to roughly 47.6% of the notional. This is plausible: each annual ATM call with 19% volatility is worth approximately 9.5% of the underlying, and five such coupons sum to about 47.5%.

At what price would ISP sell?

The theoretical price of the option is the risk-free Black-Scholes price. However, in the real market, we must consider counterparty risk.

1. The Risk Asymmetry

Bank XX buys the option and pays the premium upfront. ISP only has future payment obligations for the next 5 years. Therefore, **only Bank XX faces counterparty risk**. ISP faces zero credit risk because Bank XX will not make any future payments.

2. Bank XX's Perspective

Since ISP could default, Bank XX wants a discount on the price. This discount depends on ISP's Probability of Default. For Bank XX, the fair price is the Risk-Free Price minus this discount.

3. ISP's Perspective (The Seller)

ISP wants to maximize its profit. ISP will try to sell the option at the highest possible price. ISP will ask for the full risk-free price, plus an extra spread to cover its hedging costs and profit margins.

Conclusion

Bank XX will negotiate for a lower price ($V_{risk-free}$ − Discount). On the other hand, ISP will try to sell at a higher price ($V_{risk-free}$ + Margins). The final transaction price will depend on their negotiation.

Exercise 4: MBS Pricing

Setup

On 15/02/2008, the SPV *Cayman II* reference portfolio has €1 bn notional. The pool is homogeneous: $p = 5\%$, $\rho = 40\%$, $R = 20\%$ (LGD = 80%), single payment at $T = 3$ years. We price tranches using three methods under the Vasicek one-factor model:

- **LHP** (Large Homogeneous Portfolio): analytical limit for $I \to \infty$;
- **Exact**: finite-pool computation via binomial sum (feasible up to $I = 400$);
- **KL** (Kullback–Leibler): Stirling/saddlepoint approximation of the binomial.

The tranche loss function is $\ell(z) = \min(\max((\text{LGD} \cdot z - K_d)/(K_u - K_d), 0), 1)$, and for the inner sum in the exact method we use `binopdf` to avoid overflow in `nchoosek`.

We select 30 log-spaced points to ensure a smooth convergence plot for Point 4 while keeping the computational cost manageable.

Exercise 4a: Mezzanine Tranche [5%, 9%]

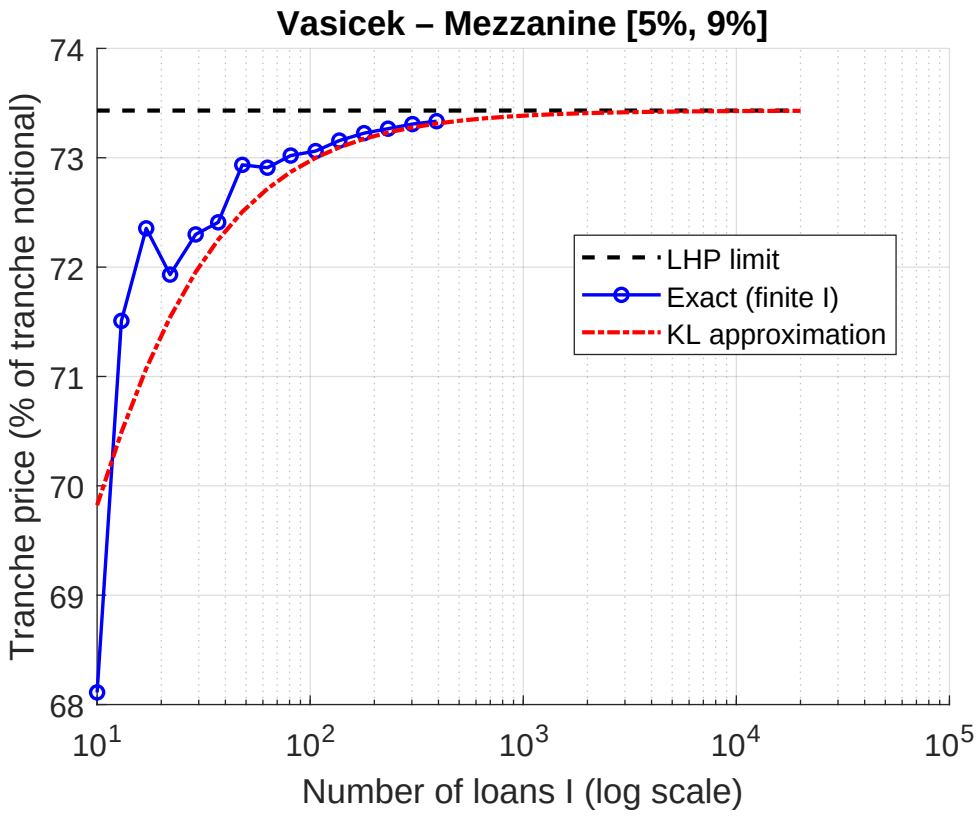

Figure 1: Mezzanine tranche [5%, 9%]: price as % of tranche face value vs. number of obligors I (log scale).

Looking at the mezzanine tranche ($K_d = 5\%$, $K_u = 9\%$), the exact prices jump in steps for small portfolios because defaults happen one by one. The KL approximation smooths out these steps perfectly. As the number of loans (I) grows, the exact price quickly reaches the Large Homogeneous Portfolio (LHP) limit. By $I = 400$, the exact price and the LHP limit are basically the same.

The tranche price is quite high, moving from 68% to about 73.4% . Why? The portfolio's average loss is 4% (a 5% default chance $\times$ 80% loss given default). Since the equity tranche takes the first 5% of losses, it absorbs all this average loss. The mezzanine tranche is safe from average losses and only drops in value in extreme scenarios.

Finally, the curve goes up as I increases because of diversification. With few loans, a couple of unlucky defaults can easily push the total loss over 5%, hurting the mezzanine tranche. With many loans, this random individual risk disappears. Total losses stay closer to the 4% average. Since 4% is below the 5% starting point of our tranche, the tranche becomes safer, and its price goes up.

Exercise 4b: Equity Tranche [0%, 5%] **(Facultative)**

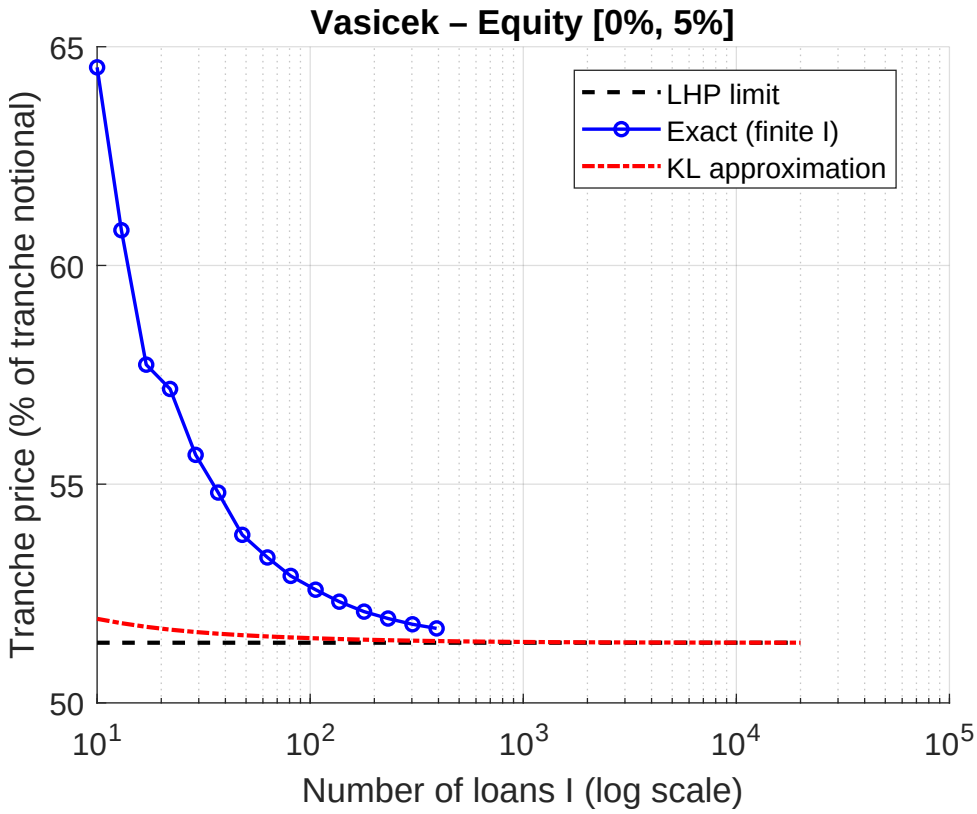

Figure 2: Equity tranche [0%, 5%]: price as % of tranche face value vs. number of obligors I (log scale).

For the equity tranche ($K_d = 0\%$, $K_u = 5\%$), we see the exact opposite trend: the price drops as I increases. As noted above, the portfolio's average loss is 4%, which falls entirely inside this tranche. With a small portfolio, there is a lucky chance of having zero defaults, keeping the tranche safe. However, as the portfolio grows, diversification guarantees that the losses will concentrate exactly around that 4% average, ensuring the equity tranche takes a severe hit and driving its price down to the LHP limit.

Is the KL approximation adequate? No. For the equity tranche, the KL approximation deviates significantly from the exact solution, especially for small I. The exact price shows a steep drop from ∼65% at $I = 10$ to ∼51% at $I = 400$ and converges to the LHP limit, while the KL curve does not capture this behaviour well.

The reason is structural: the equity tranche has $K_d = 0$, so the tranche loss function is non-zero starting from $z = 0$. The KL approximation relies on the Stirling prefactor $C_1(z) = \sqrt{I}/\sqrt{2\pi z(1-z)}$, which diverges as $z \to 0$. This makes the saddlepoint approximation unreliable precisely in the region that matters most for the equity tranche.

How can you modify it? A hybrid approach can be used: treat the first few default counts ($m = 0, 1, 2, \ldots$) exactly via the binomial formula, and apply the KL continuous approximation only for $m \geq m_{\min}$ where the Stirling approximation is accurate. This avoids the divergence near $z = 0$ while retaining the computational efficiency of the KL method for the bulk of the distribution.

Additional Functions Implemented

- `[dates, discounts, zeroRates] = bootstrap(datesSet, ratesSet)`

- `[ES, VaR] = BootstrapMeasures(alpha, weights, portfolioValue, ... riskMeasureTimeIntervalInDays, returns, M)`
- `adjusted_date = business_date_offset(base_date, varargin)`
- `[newDate, idx] = closestDate(date, dateVec, adjRule)`
- `[VaR_PCA, ES_PCA, relErrorVaR, k_5, k_1] = compute_PCA_and_errors(alpha, weights, portfolioValue, riskMeasureTimeIntervalInDays, returnsSelected)`
- `priceCliquet = FairPriceCliquet(S0, r, sigma)`
- `[disc_fact, rate] = find_rate_from_bootstrap_curve(refDate, maturity)`
- `zero_rates = from_discount_factors_to_zero_rates(reference_date, dates, discounts)`
- `discount_interp = get_discount_factor_by_zero_rates_linear_interp(reference_date, interp_date, dates, discounts)`
- `val = innerSumFinite(I, py, trancheLoss)`
- `plot_tranche_prices(I_grid, price_LHP, price_exact, price_KL, notional, tranche_name, filename)`
- `[VaR_FMC, VaR_DN] = computeVaR_Exercise2(stockNotional, strike, expiryDate, refDate, alpha, volatility, dividendYield, riskMeasureTimeIntervalInDays, timeWindow, sharesList, inputFile, formatDate)`
- `[price_LHP, price_exact, price_KL] = compute_tranche_prices(I_grid, p, rho, LGD, Kd, Ku, valuationDate, expiry, I_max_exact, notional)`
- `price = tranchePriceKL_Vasicek(I, p, rho, LGD, Kd, Ku, disc_fact, notional)`
- `price = tranchePriceLHP_Vasicek(p, rho, LGD, Kd, Ku, disc_fact, notional)`
- `price = tranchePriceExact_Vasicek(I, p, rho, LGD, Kd, Ku, disc_fact, notional)`

Problems and Criticalities

Ex.2: Historical Simulation for Delta-Normal

Despite the traditional name "Delta-Normal", we understood from risk management theory that "Delta" simply refers to the first-order linearization of the portfolio. Therefore, we evaluated this linear model using actual historical returns instead of a Gaussian assumption. This choice strictly follows the assignment's explicit instruction to "Use a 2y Historical Simulation for the underlying".

From a theoretical perspective, a VaR model has two main components: the pricing function and the scenario distribution. By keeping the scenario distribution identical (Historical Simulation) for both methods, we perfectly isolate the pricing effect. This ensures that any difference between the two VaRs is strictly due to the linear approximation, allowing for a rigorous, apples-to-apples comparison.